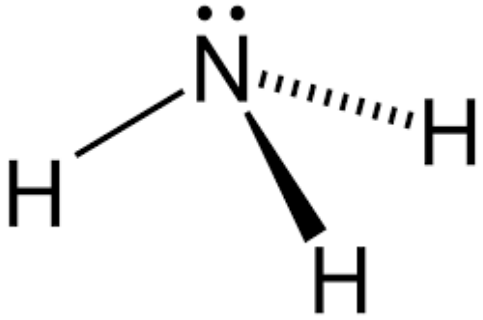
AMMONIA TOXICITY

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Ammonia

- Is a chemical compound of nitrogen and hydrogen atoms with formula of NH_3, always dissolve in water found as NH4OH



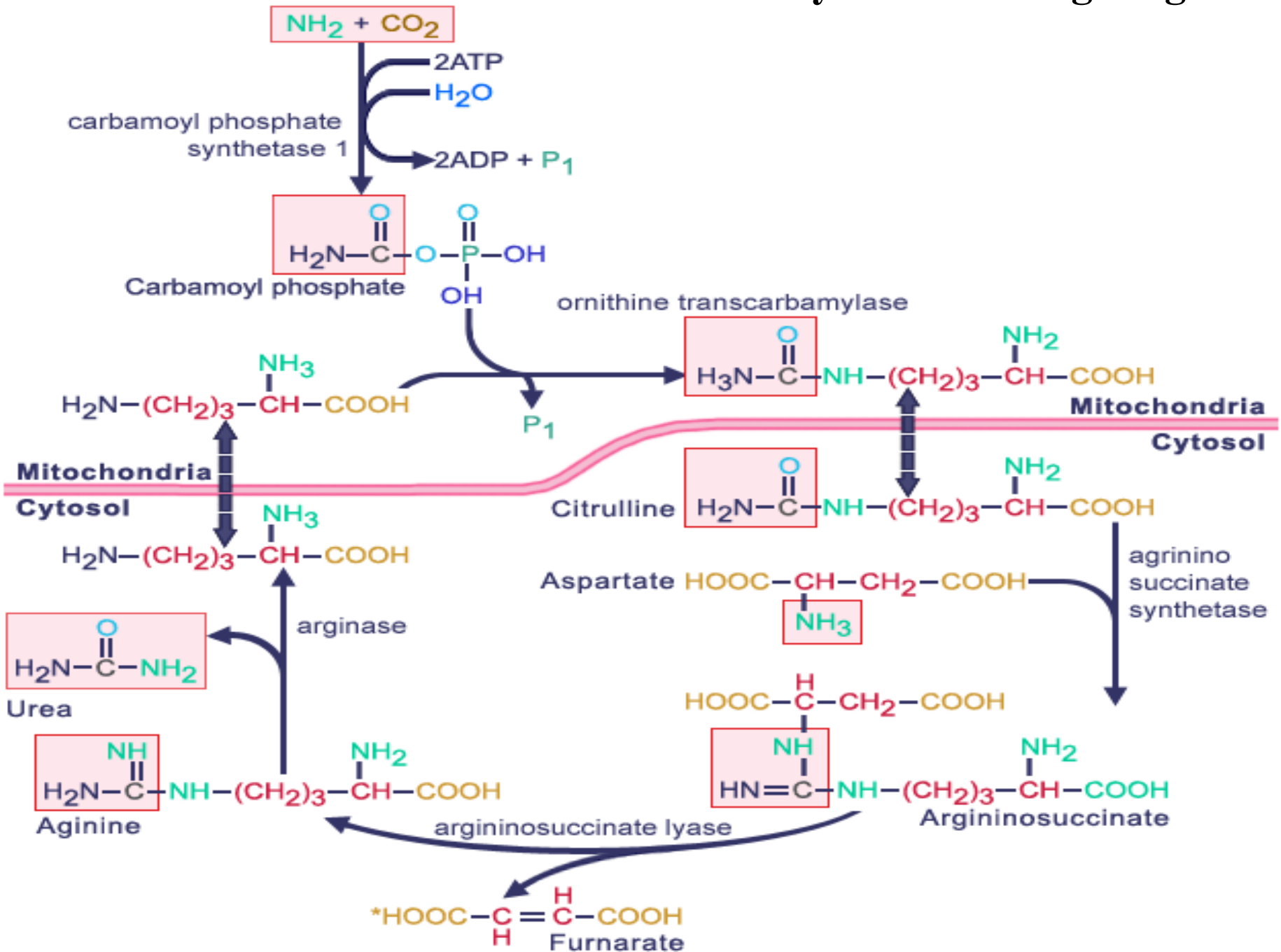
Some facts about ammonia

- Ammonia is normally present in all body fluids including CSF in CNS
- The normal blood ammonium is normally lesser than 50 micro mol\L (85 ug/dL)
- When ammonia is accumulated in abnormal quantities leads to hyperammonemia
- The major source of ammonia is amino acid metabolism.
- High ammonia level may be a sign of underlying cause.

So HOW body regulate the normal amount of ammonia in order to preserve homeostasis ?and avoid toxicity

Ammonia is a toxic product of nitrogen metabolism which should be removed from our body. The urea cycle or ornithine cycle converts excess ammonia into urea in the mitochondria and cytoplasm of liver cells. The urea forms, then enters the blood stream, is filtered by the kidneys and is ultimately excreted in the urine.

These reactions can be illustrated by the following diagram



:The causes of hyperammonemia

- Primary
  (hereditary urea cycle enzyme defects)

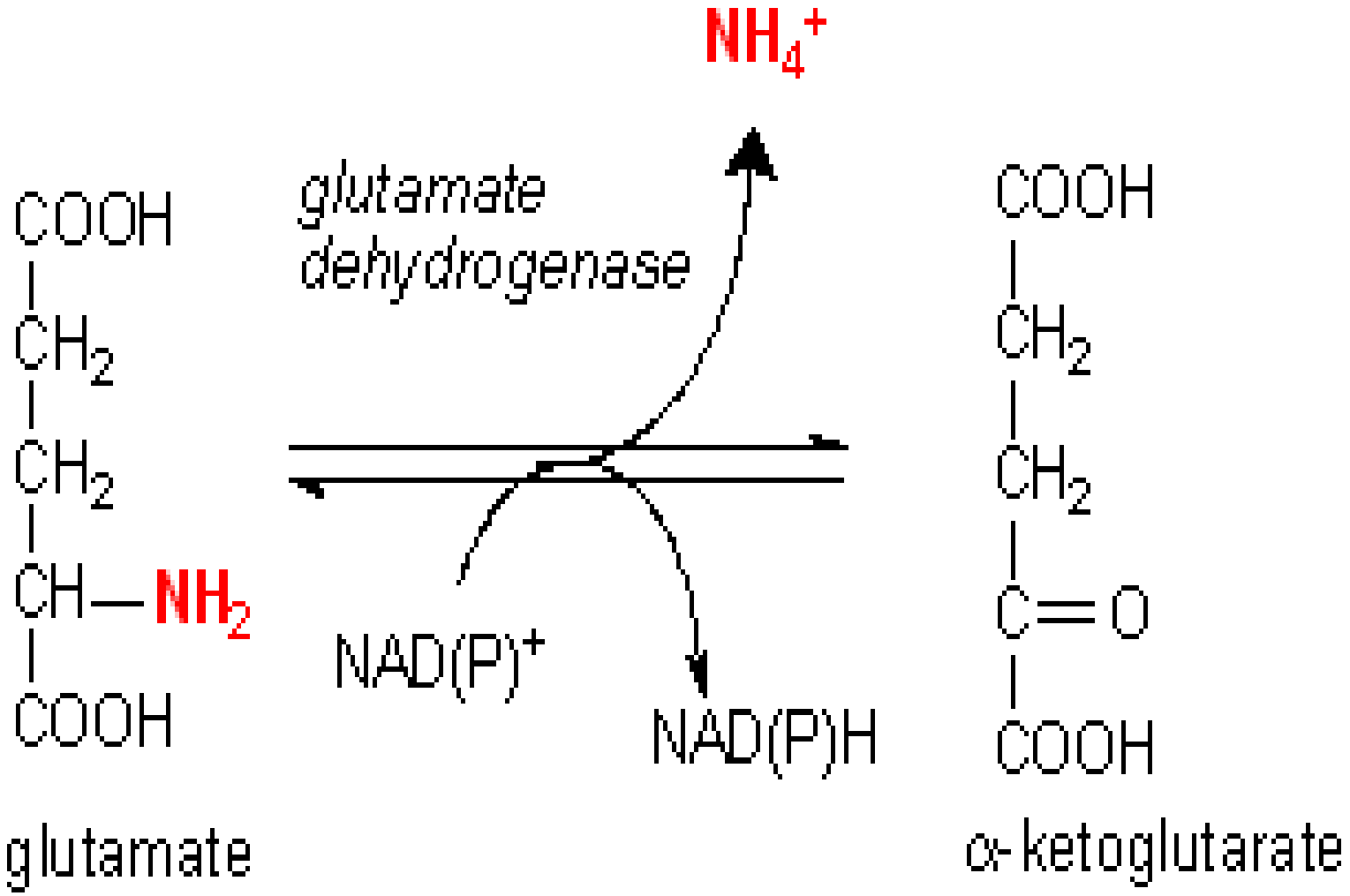
- Secondary
  (acquired: liver damage)

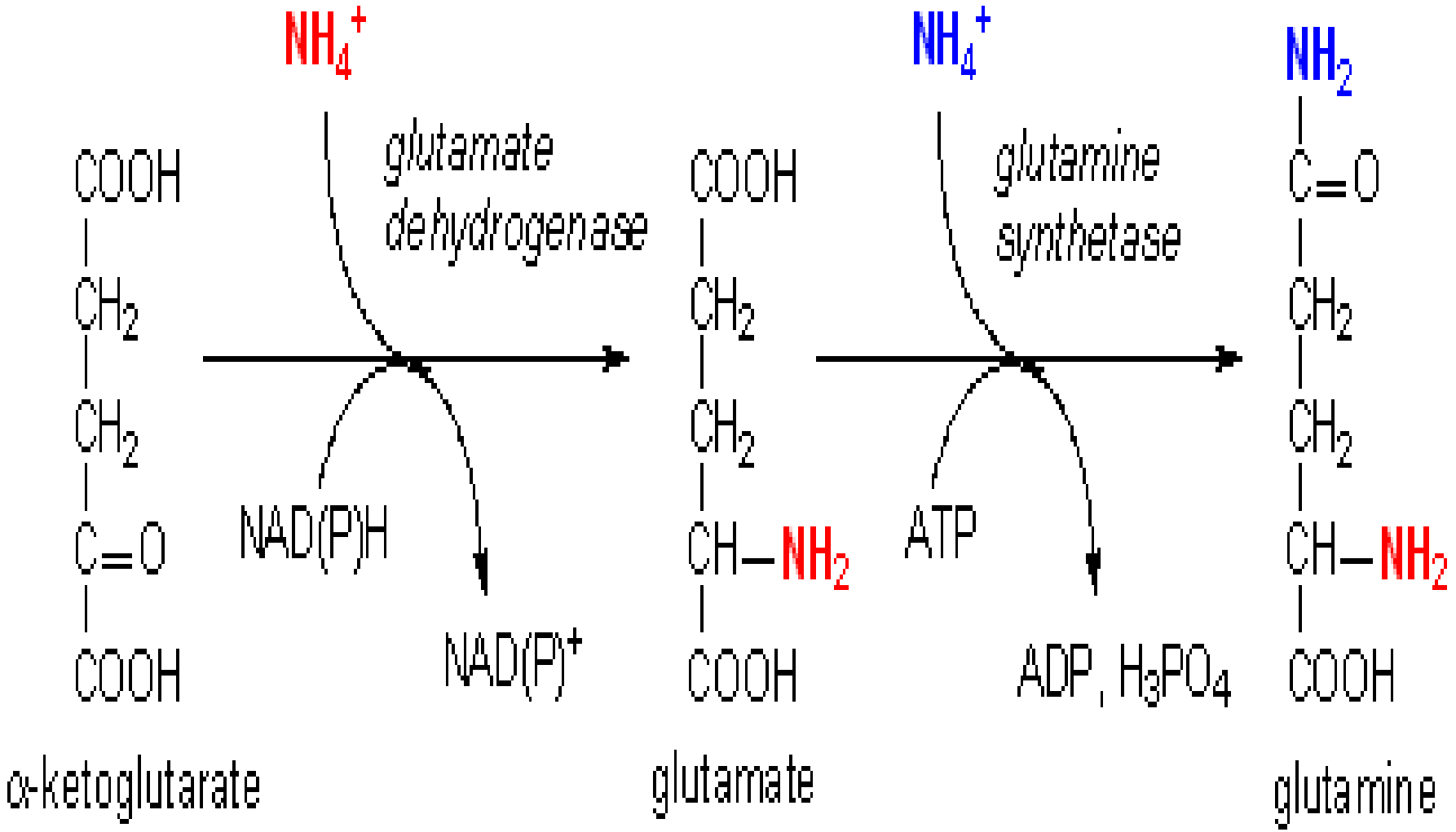
Acquired cause: LIVER DAMAGE

- High protein diet
- Liver cirrhosis
- Infections (hepatitis)
- Liver injury
- Medications
- Alcohol
- Processed food
- High insulin

METABOLISM OF GLUTAMATE

- In body, normally alpha-ketoglutarate binds with ammonium under the catalysis of glutamate dehydrogenase forming glutamate.
- Glutamate bind with ammonium under catalysis of glutamine synthetase forming glutamine.
- Alpha-ketoglutarate is an important compound in krebs cycle, hence ATP production.


NH4+
glutamate
dehydrogenase
COOH
CH2
CH2
CH—NH2
COOH
NAD(P)+
NAD(P)H
COOH
CH2
CH2
C=O
COOH
glutamate
α-ketoglutarate

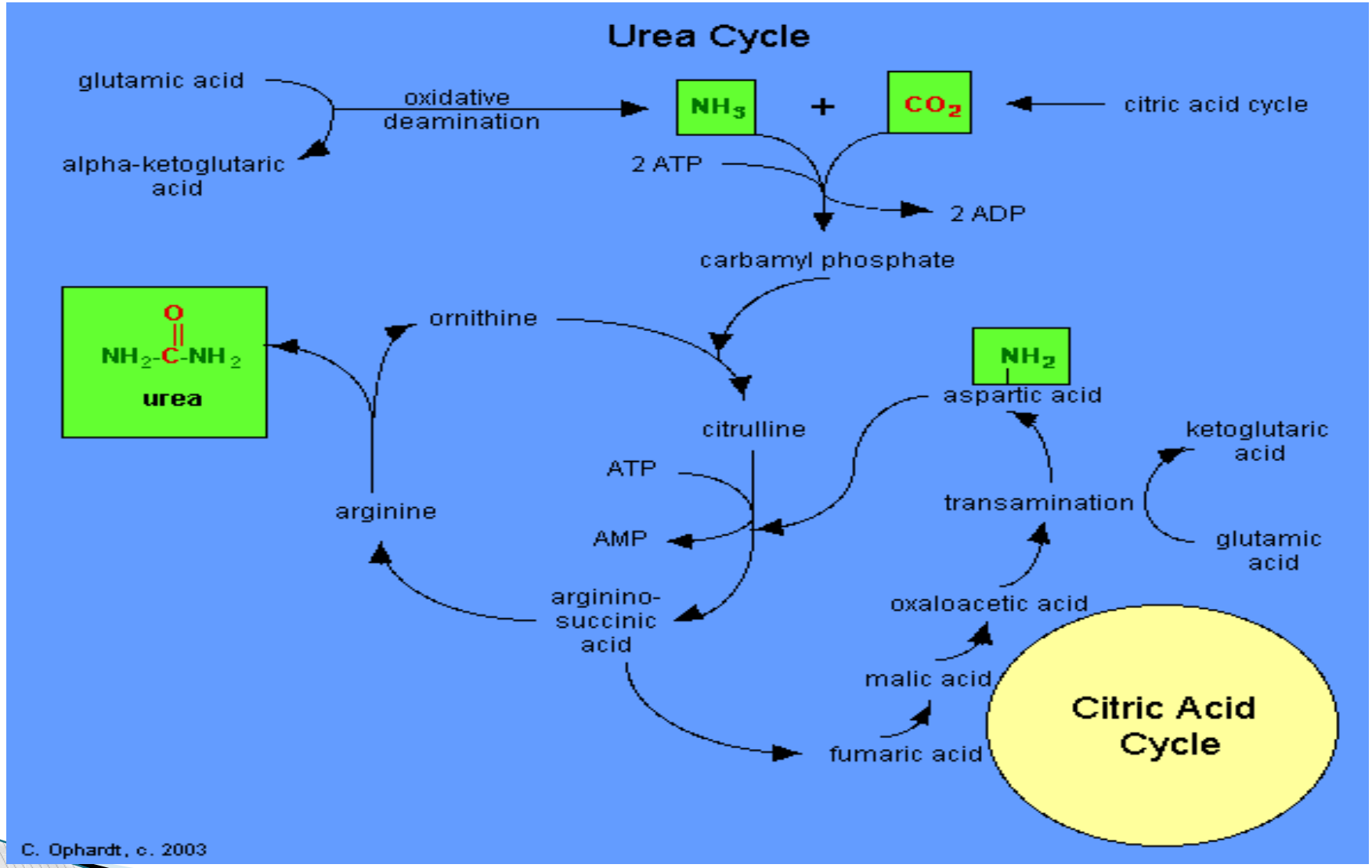

NH4+
glutamate dehydrogenase
NAD(P)H
NAD(P)+
COOH
CH2
CH2
C=O
COOH
α-ketoglutarate
NH4+
glutamine synthetase
ATP
ADP, H3PO4
COOH
CH2
CH2
CH—NH2
COOH
glutamate
NH2
C=O
CH2
CH2
CH—NH2
COOH
glutamine

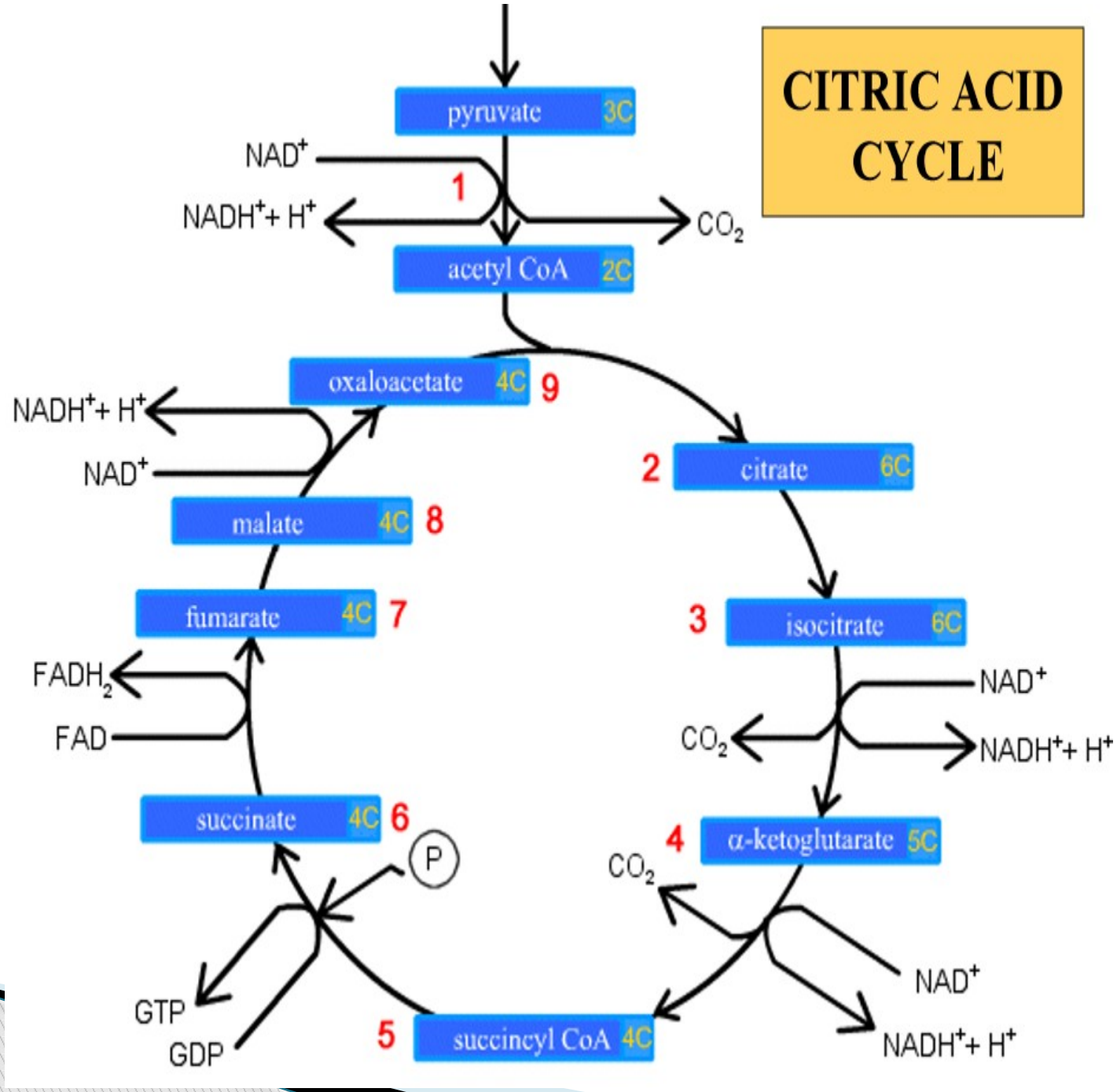
The toxic effects of hyperammonemia on body :(especially BRAIN)

- Inhibits ATP production by inhibiting Krebs cycle, that affects ion transport across nerve cell membranes, so resulting in disturbance, then loss, of consciousness.
- In glutamate metabolism, ammonia makes the production of glutamine dominant
- SO, low levels of alpha-ketoglutarate that inhibits Krebs cycle
- And low levels of glutamate which is the precursor of GABA (inhibitory neurotransmitter).

Symptoms of ammonia toxicity

- Toxic to brain since it alters glial cells ability to remove **potassium.**
- It may cause seizures, coma, and can even lead to death.


Urea Cycle
glutamic acid
oxidative deamination
alpha-ketoglutaric acid
NH3
+
CO2
citric acid cycle
2 ATP
2 ADP
carbamyl phosphate
ornithine
NH2-C-NH2
urea
citrulline
NH2
aspartic acid
ketoglutaric acid
ATP
AMP
arginine
transamination
glutamic acid
arginino-succinic acid
oxaloacetic acid
malic acid
fumaric acid
Citric Acid Cycle
C. Ophardt, c. 2003


CITRIC ACID CYCLE
pyruvate 3C
NAD+
1
NADH+ H+
CO2
acetyl CoA 2C
oxaloacetate 4C 9
NADH+ H+
NAD+
2 citrate 6C
malate 4C 8
fumarate 4C 7
3 isocitrate 6C
FADH2
FAD
NAD+
CO2
NADH+ H+
succinate 4C 6
P
4 α-ketoglutarate 5C
CO2
GTP
GDP
5 succincyl CoA 4C
NAD+
NADH+ H+

Normal Range Of Blood Ammonia

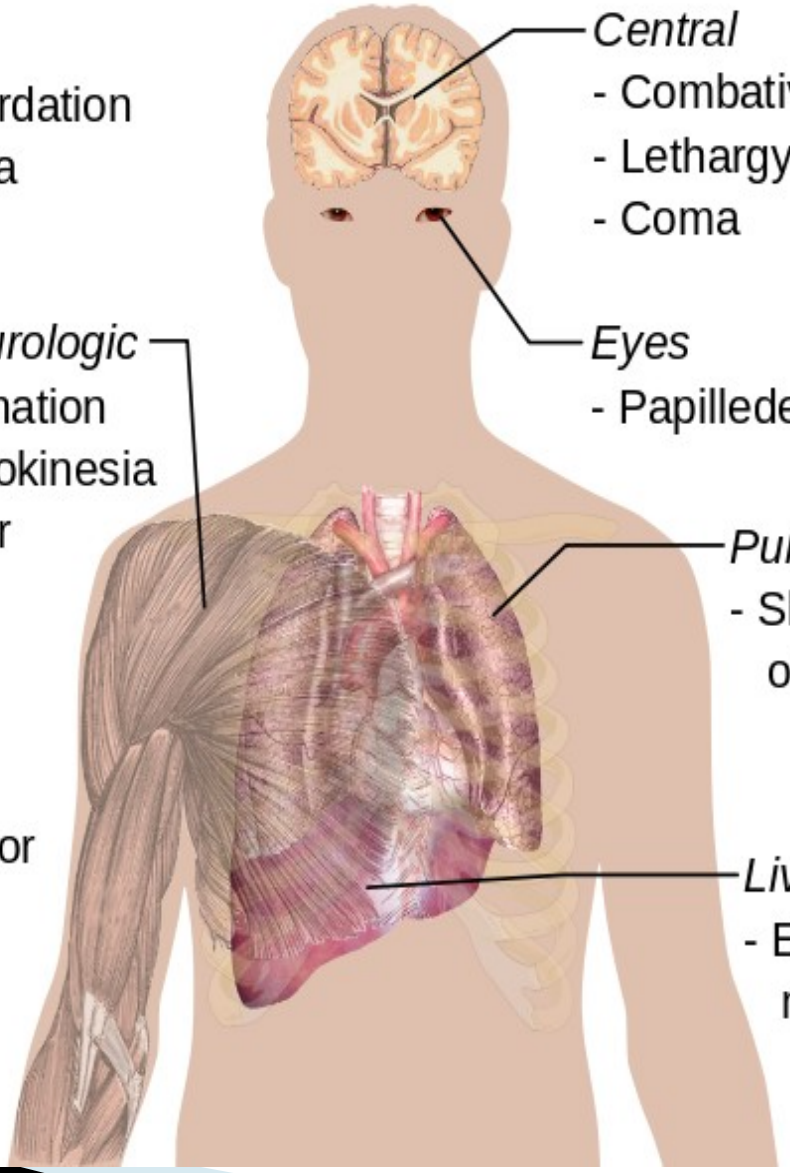
The normal amount of ammonia found in the blood is about 50µmol/L. If the level of ammonia reaches 100µmol/L, the patient may experience alterations in their level of consciousness. An ammonia level of 200µmol/L may lead to convulsions and coma.

:Clinical features of hyperammonemia

- Confusion
- Memory loss
- Swelling in the brain (edema)
- Musty sweat & breath
- Vomiting (>24 hrs)
- Overly sleeping
- encephalopathy
- Problems in thinking
- Problems with balance
- Tremor
- Slurring of speech
- Blurry vision
- Convulsions and seizure

Symptoms of

Hyperammonemia



General
- Growth retardation
- Hypothermia

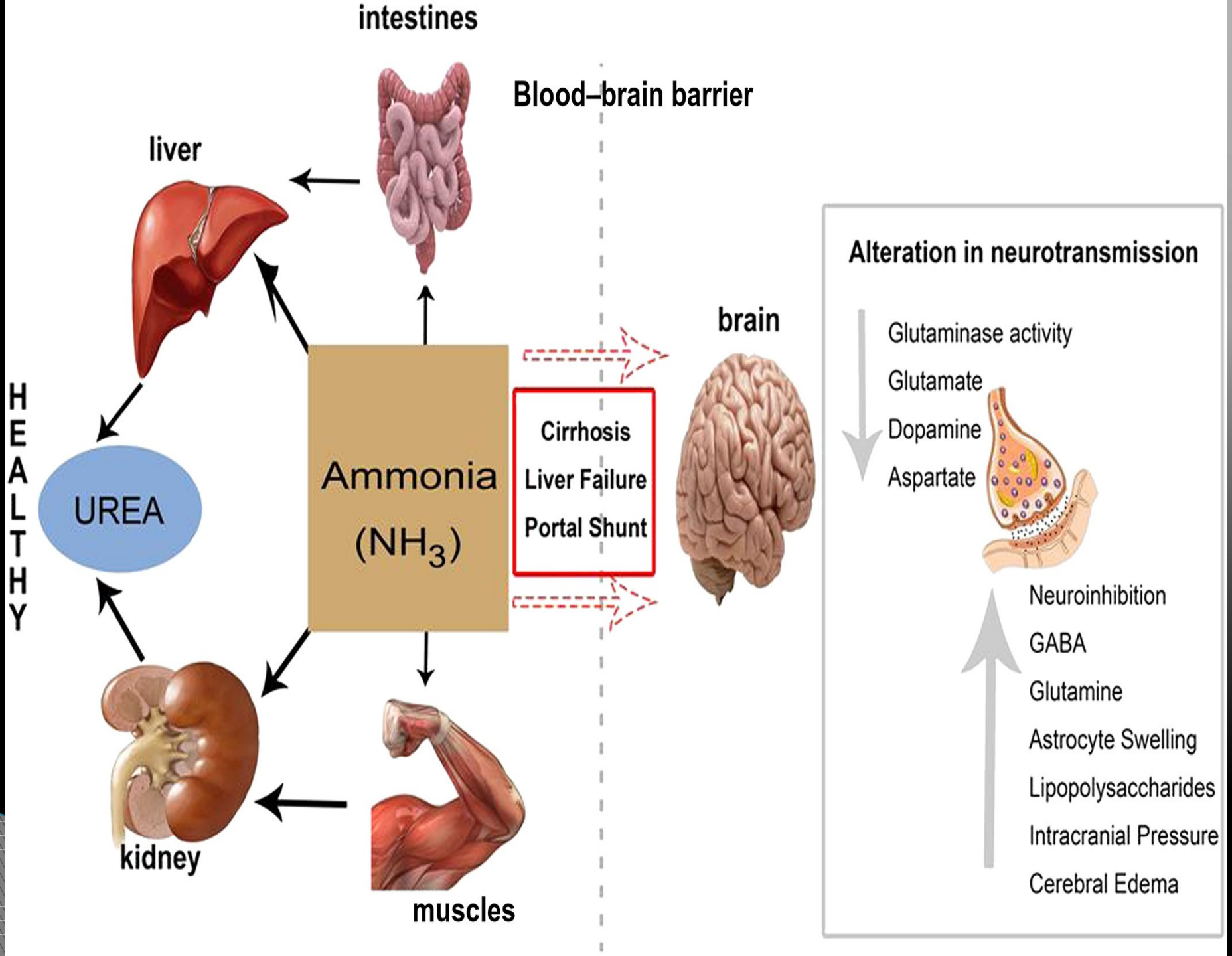
Central
- Combativeness
- Lethargy
- Coma

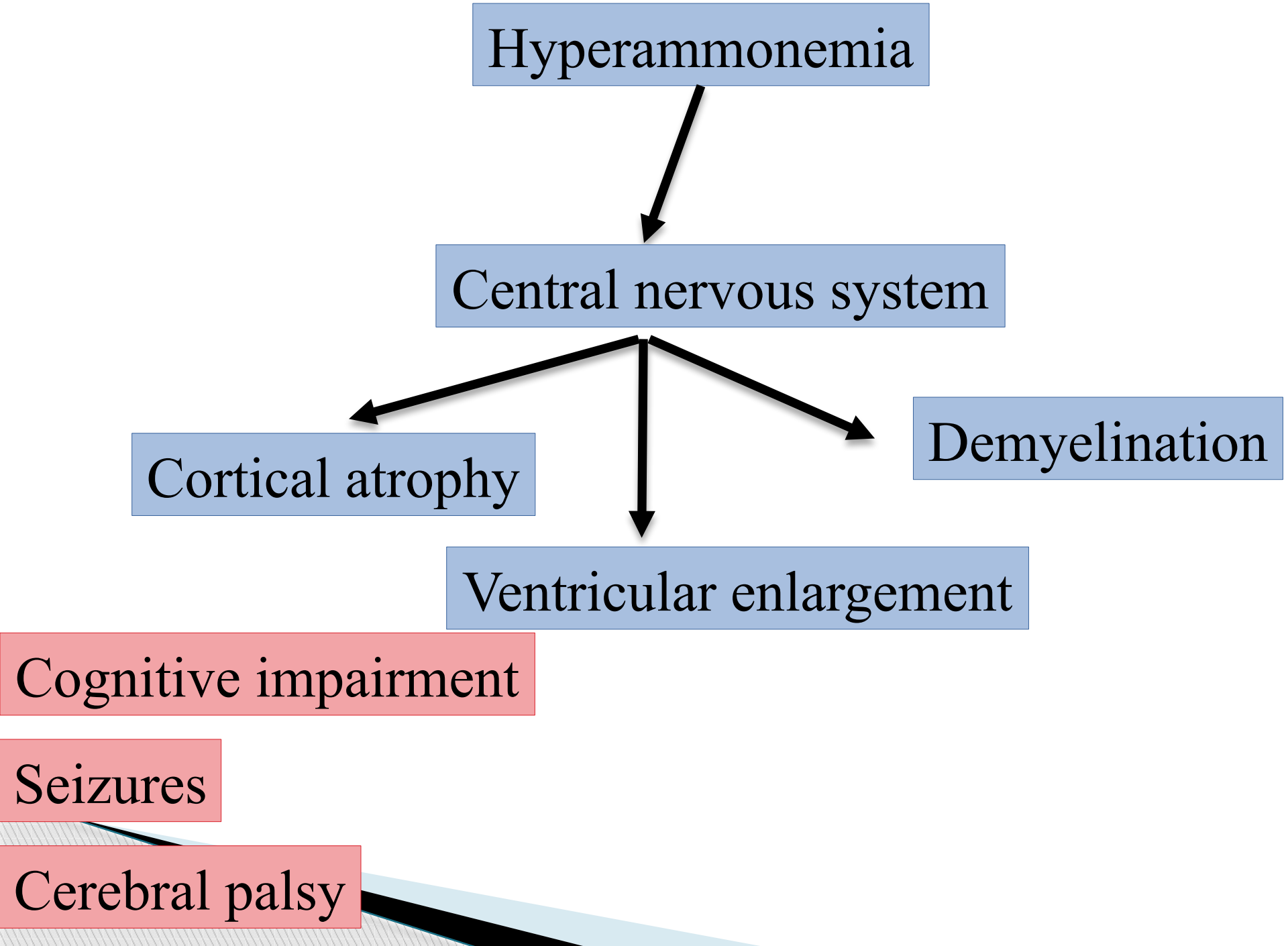
Eyes
- Papilledema

Muscular/Neurologic
- Poor coordination
- Dysdiadochokinesia
- Hypotonia or hypertonia
- Ataxia
- Tremor
- Seizures
- Decorticate or decerebrate posturing

Pulmonary
- Shortness of breath

Liver
- Enlarge-ment


intestines
Blood–brain barrier
liver
brain
HEALTHY
UREA
Ammonia
(NH_3)
Cirrhosis
Liver Failure
Portal Shunt
kidney
muscles
Alteration in neurotransmission
Glutaminase activity
Glutamate
Dopamine
Aspartate
Neuroinhibition
GABA
Glutamine
Astrocyte Swelling
Lipopolysaccharides
Intracranial Pressure
Cerebral Edema


Hyperammonemia
Central nervous system
Cortical atrophy
Demyelination
Ventricular enlargement
Cognitive impairment
Seizures
Cerebral palsy

Intracranial hypertension

ed by ammonia toxicity in brain

Memory impairment

Shortened attention span

Sleep-wake inversion

So how we will protect ourselves from hyperammonemia or what are the treatments ?

Protect your LIVER

- Avoid toxins (alcohol, drugs…)
- Discuss all your medications or supplements with your doctor
- Limit protein in diet
- Healthy weight (diet, exercise)
- Regular water drinking
- Fruits (banana)
- Vegetables
- Fish
- Garlic (it activates the enzymes that have role in intoxification).


Chiquita
Chiquita
Chiquita

:Treatment

- **Enzymatic deficiency→ Sodium Banzoate and sodium phenylbutyrate (adjunctive, help ammonia excretion).**

- **Severe hyperammonemia (>1000 micro mol/L)→ begin with hemodialysis (purifying the blood).**

Treatment of ammonia toxicity

:Burinex

Diuretic drug bumetanide

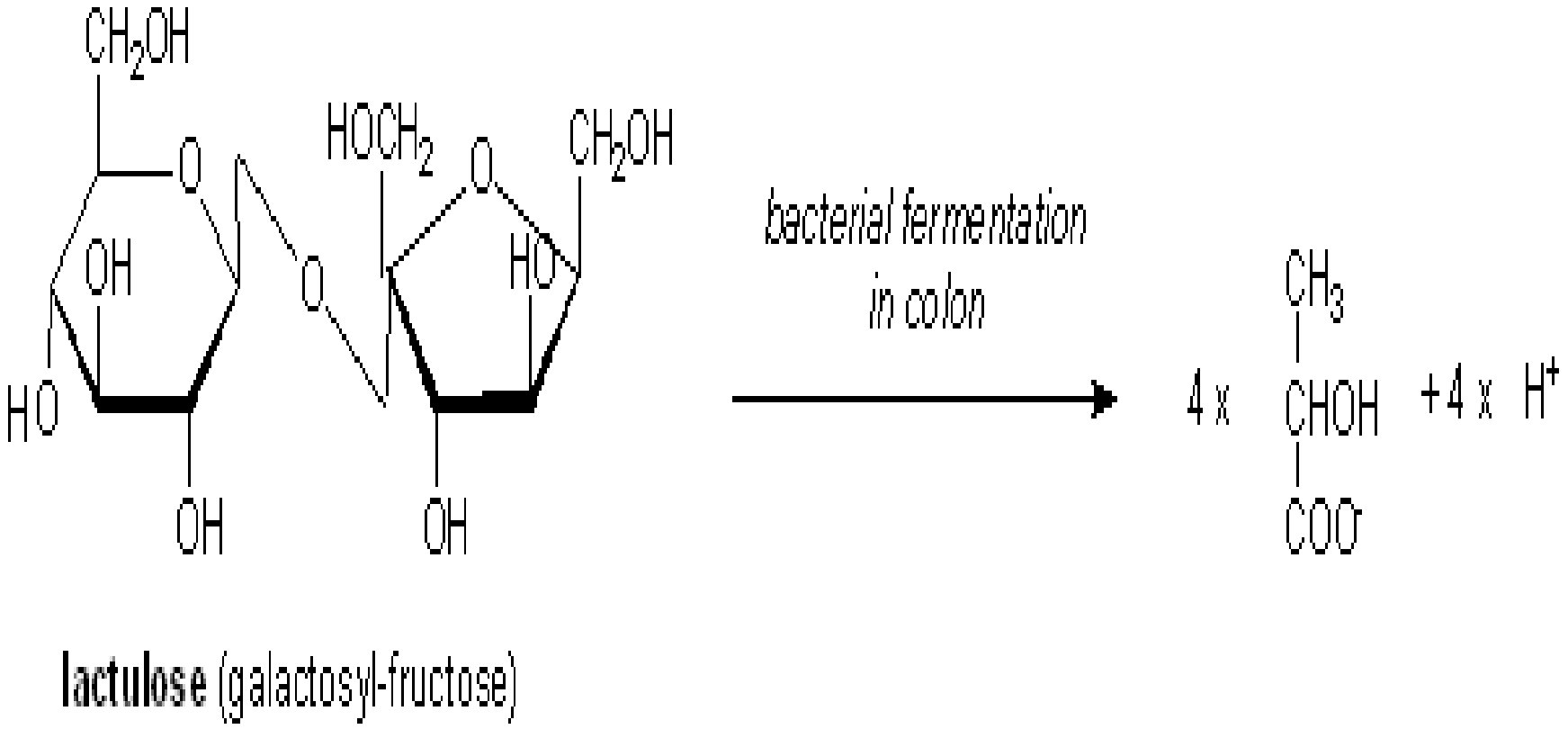


Bumetanide block the transport of **potassium ions** into the nerve cells,

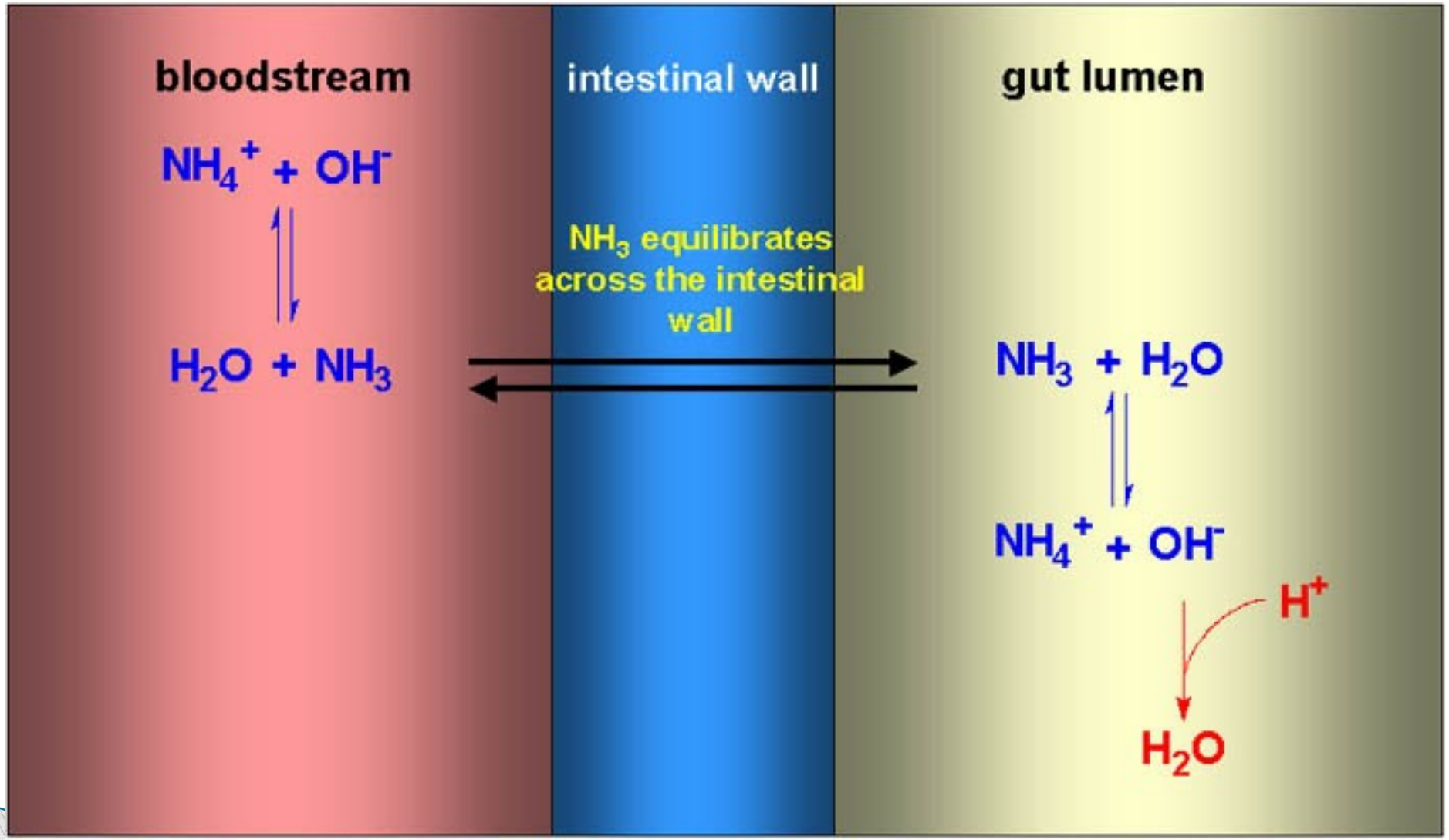
Thereby protecting them from harm.

Hepatic encephalopathy in adults, or sometimes, in infants whose blood ammonium has risen dangerously high (commonly as a result of a genetic disease affecting amino acid metabolism →Lactulose

lactulose (synthetic, by gut flora fragmentation convert into lactate and producing H+ which binds with ammonia forming ammonium ions NH_4^+ that can not be absorbed due to changes in pH or acidification in a process called "ammonium trapping".



lactulose (galactosyl-fructose)


bloodstream
intestinal wall
gut lumen
NH4+ + OH-
NH3 equilibrates across the intestinal wall
H2O + NH3
NH3 + H2O
NH4+ + OH-
H+
H2O


Thank you!